

9.5: The Ratio, Root, and Comparison Tests.

Calculus III | MTH 331 | Section A | Fall 2012

Dr. James Ashe

Johnson C. Smith University

../../jcsu_bull.jpg

Dr. James Ashe

Short Title

The Ratio Test

Theorem. The Ratio Test

Given a series $\sum a_k$, let $r = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right|$.

- 1 If $0 \leq r < 1$ the series converges.
- 2 If $1 < r$ then the series diverges.
- 3 If $r = 1$ the test is inconclusive. Use

Use the Ratio Test to determine whether the following series converge.

ex 9. $\sum_{k=1}^{\infty} \frac{1}{k!}$.

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} &= \lim_{k \rightarrow \infty} \frac{1/(k+1)!}{1/k!} = \lim_{k \rightarrow \infty} \frac{k!}{(k+1)!} \\ &= \lim_{k \rightarrow \infty} \frac{1}{1 \cdot 2 \cdot 3 \cdots k \cdot (k+1)} = \lim_{k \rightarrow \infty} \frac{1}{(k+1)} = 0. \Rightarrow \sum_{k=1}^{\infty} \frac{1}{k!} \text{ converges.} \end{aligned}$$

Dr. James Ashe

Short Title

The Ratio Test

Why does the Ratio Test work?

- As k gets large, $r = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| \Rightarrow r \cdot a_k \approx a_{k+1}$.
- It is the "tail" of the series that determines its end behavior.
- So then $a_k + a_{k+1} + a_{k+2} + \cdots \approx a_k + r \cdot a_k + r^2 \cdot a_k + \cdots \approx a_k(1 + r + r^2 + \cdots)$ behaves geometrically;
geometric series converge when $r < 1$ and diverge when $r \geq 1$.

Dr. James Ashe

Short Title

The Ratio Test

ex. $\sum_{n=1}^{\infty} \frac{9^n}{(-2)^{n+1}n}$.

- Note: $\frac{a_{n+1}}{a_n} = a_{n+1} \left(\frac{1}{a_n} \right)$

$$\lim_{n \rightarrow \infty} \left[a_{n+1} \left(\frac{1}{a_n} \right) \right] = \lim_{n \rightarrow \infty} \left[\frac{9^{n+1}}{(-2)^{n+2}(n+1)} \cdot \frac{(-2)^{n+1}n}{9^n} \right] =$$

$$\lim_{n \rightarrow \infty} \frac{9n}{(-2)^1(n+1)} = -\frac{9}{2} \lim_{n \rightarrow \infty} \frac{n}{(n+1)} = -\frac{9}{2}$$

$$, \text{ and } \left| -\frac{9}{2} \right| > 1 \Rightarrow \sum_{n=1}^{\infty} \frac{9^n}{(-2)^{n+1}n} \text{ diverges.}$$

Dr. James Ashe

Short Title

The Root Test

Theorem: The Root Test

Let $\sum a_k$ be an infinite series with *nonnegative* terms and let $\rho = \lim_{k \rightarrow \infty} \sqrt[k]{a_k}$,

- ① If $0 \leq \rho < 1$, then the series converges.
- ② If $1 < \rho$, then the series diverges.
- ③ If $\rho = 1$, the test is inconclusive.

ex 20. Evaluate $\sum_{k=1}^{\infty} \left(\frac{k+1}{2k} \right)^k$.

$$\sqrt[k]{\left(\frac{k+1}{2k} \right)^k} = \left(\frac{k+1}{2k} \right). \quad \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \lim_{k \rightarrow \infty} \frac{n/n + 1/n}{2n/n} = \frac{1+0}{2} = \frac{1}{2}.$$

The Root Test

Why does the Root Test work?

- As k gets large, $\rho \approx \sqrt[k]{a_k}$ or $a_k \approx \rho^k$.
- Again, it is the “tail” of the series that determines its end behavior.
- So then $a_k + a_{k+1} + a_{k+2} + \dots \approx \rho^k + \rho^{k+1} + \rho^{k+2} + \dots$ behaves geometrically;
geometric series converge when $\rho < 1$ and diverge when $r \geq \rho$.

The Comparison Test

Theorem. Comparison Test

Let $\sum a_k$ and $\sum b_k$ be series with positive terms.

- ① If $0 < a_k \leq b_k$ and $\sum b_k$ converges then $\sum a_k$ also converges.
- ② If $0 < b_k \leq a_k$ and $\sum b_k$ diverges then $\sum a_k$ also diverges.

ex 31. Evaluate $\sum_{k=1}^{\infty} \frac{k}{k^2 - \cos^2(k)}$.

$$0 \leq \cos^2(x) \leq 1$$

$\frac{k}{k^2 - \cos^2(k)} > \frac{k}{k^2} = \frac{1}{k}$ (making the denominator smaller $\Rightarrow$ fraction gets bigger)

$\sum_{k=1}^{\infty} \frac{1}{k} = \infty$ (the Harmonic Series) so then $\sum_{k=1}^{\infty} \frac{k}{k^2 - \cos^2(k)}$ also diverges.

The Limit Comparison Test

Theorem. Limit Comparison Test

Define $L = \lim_{k \rightarrow \infty} \frac{a_k}{b_k}$ where $a_n, b_n \geq 0$ for all n .

- ① If $0 < L < \infty$ then $\sum a_k, \sum b_k$ either both converge or diverge.
- ② If $L = 0$ and $\sum b_k$ converges then $\sum a_k$ converges.
- ③ If $L = \infty$ and $\sum a_k$ diverges then $\sum b_k$ diverges.

ex. Evaluate $\sum_{k=0}^{\infty} \frac{1}{3^k - k}$. guess: it converges

Need a series to compare (want it to converge): $\sum_{k=0}^{\infty} \frac{1}{3^k}$.

$$\lim_{k \rightarrow \infty} \frac{1/3^k}{1/(3^k - k)} = \lim_{k \rightarrow \infty} \left(1 - \frac{k}{3^k} \right) = 1 \Rightarrow \sum_{k=0}^{\infty} \frac{1}{3^k - k} \text{ converges}$$

since $\sum_{k=0}^{\infty} \frac{1}{3^k}$ does.

Homework so far

9.1: 21, 55, 60, 66, 68

9.2: 4, 9, 18, 20, 29, 46, 48, 51-56, 58

9.3: 12, 19, 23, 26, 47, 49

9.4: 12, 14, 20, 24, 26, 29

9.5: 9-14, 19-25, 27-28

Recall: Integration by Parts

$$\int u dv = uv - \int v du$$

ex. Find $\int xe^{-x} dx$. Let $u = x$ so that,

$$\begin{aligned} u &= x & dv &= e^{-x} dx \\ du &= 1 & v &= -e^{-x} \end{aligned}$$

$$\begin{aligned} \int \underbrace{x}_u \cdot \underbrace{e^{-x} dx}_{dv} &= \underbrace{x}_u \underbrace{(-e^{-x})}_v - \int \underbrace{(-e^{-x})}_v \underbrace{dx}_{du} \\ &= -xe^{-x} - e^{-x} + C \end{aligned}$$

e^x

$$\lim_{n \rightarrow \infty} \left(\frac{n+x}{n} \right)^n = e^x$$

In particular, $\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = e^{-1}$.

These facts come in useful on problems 9.4: 29, 9.5: 12 and 14 (among others)